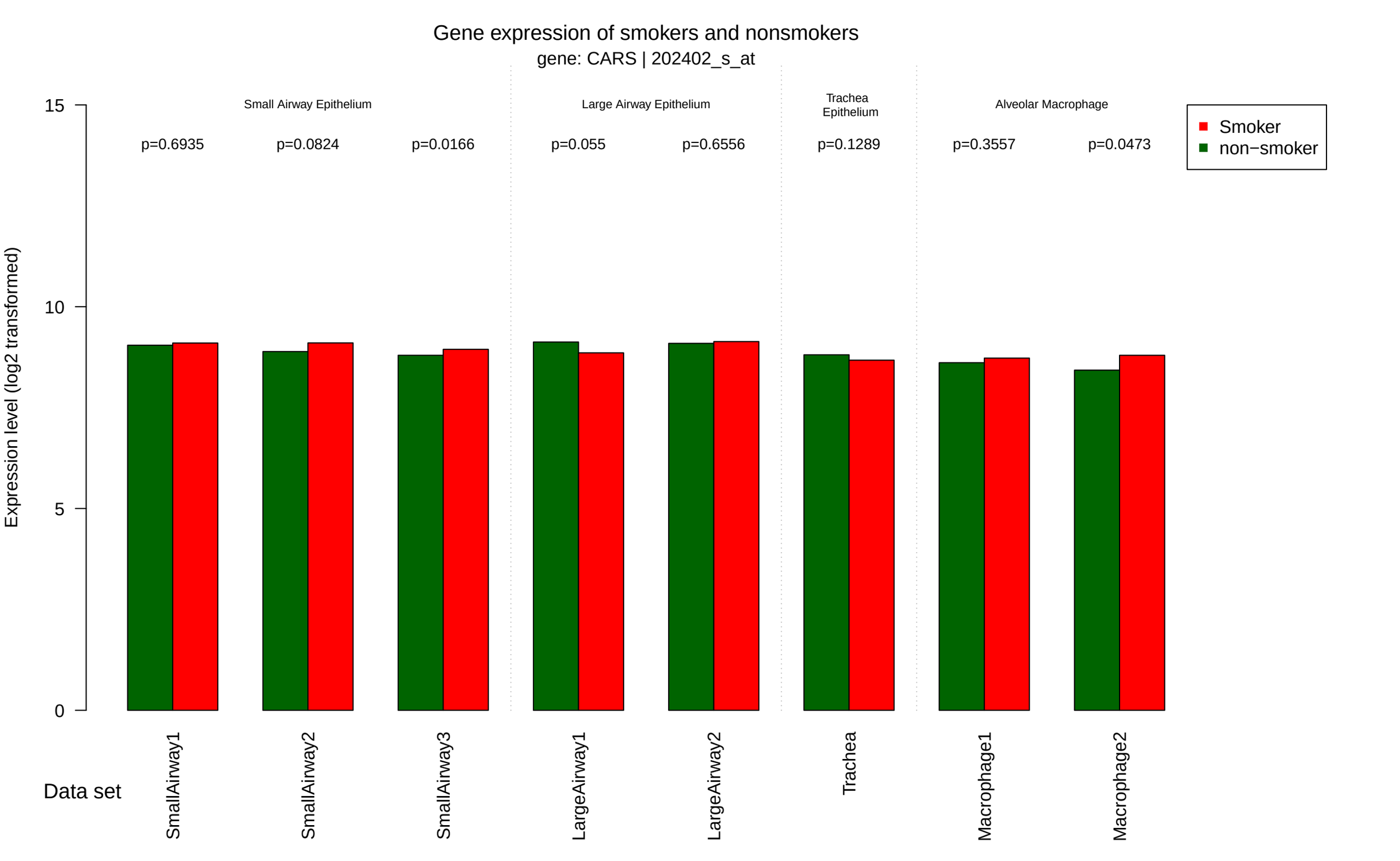
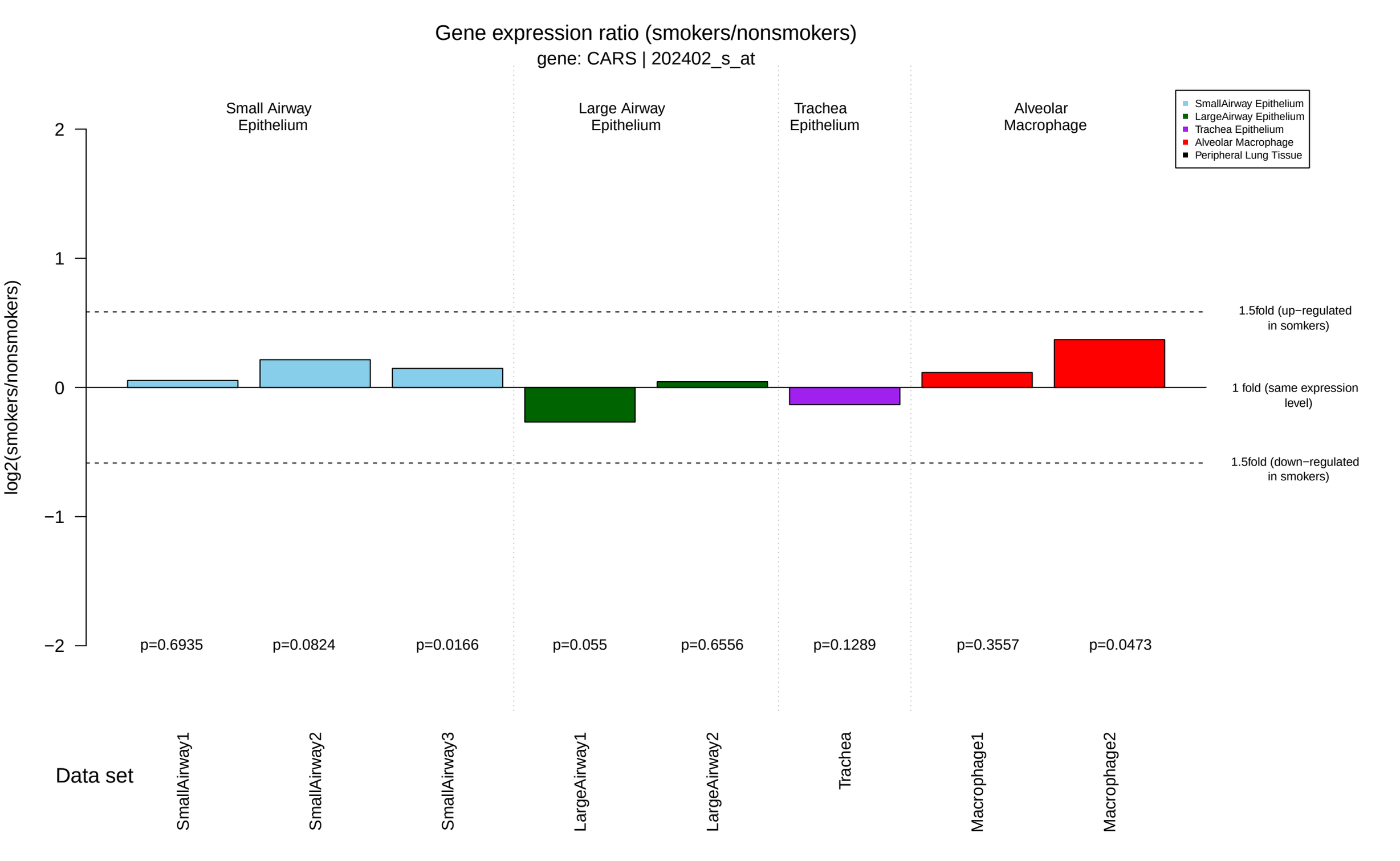
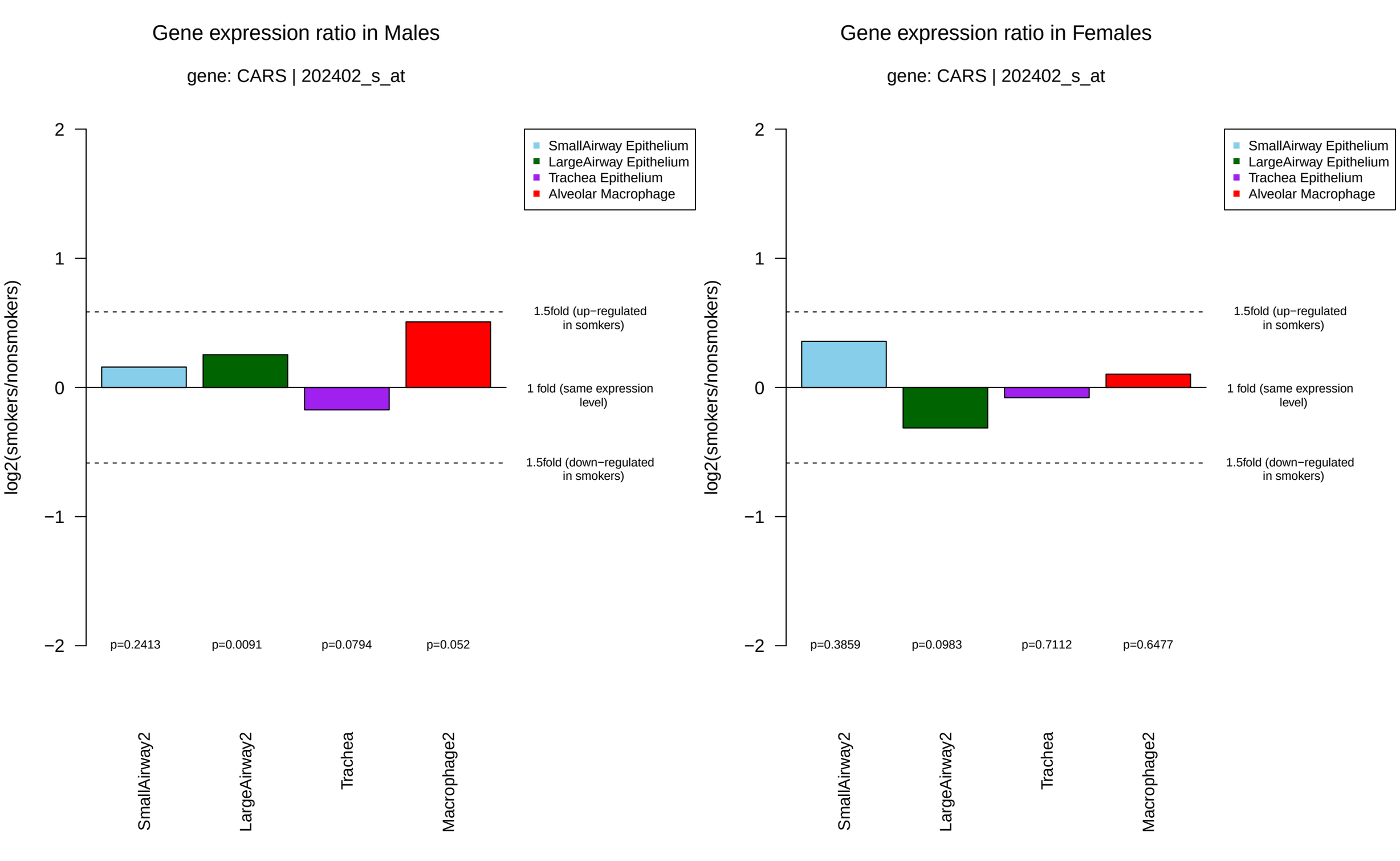
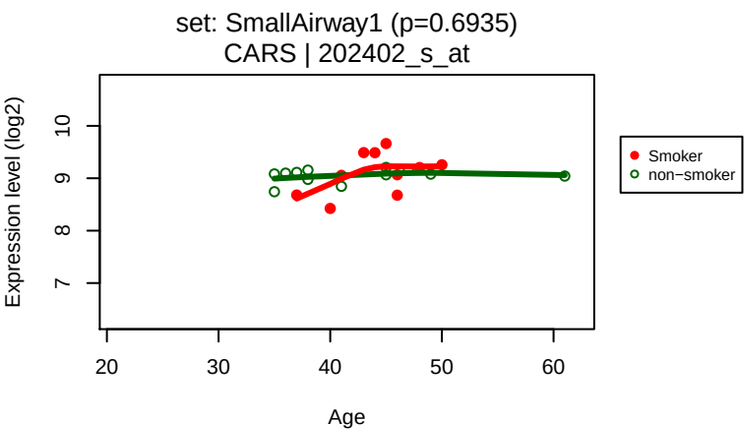
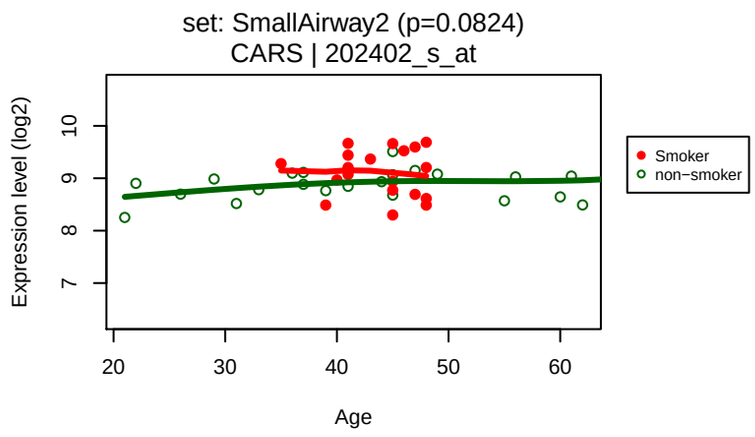
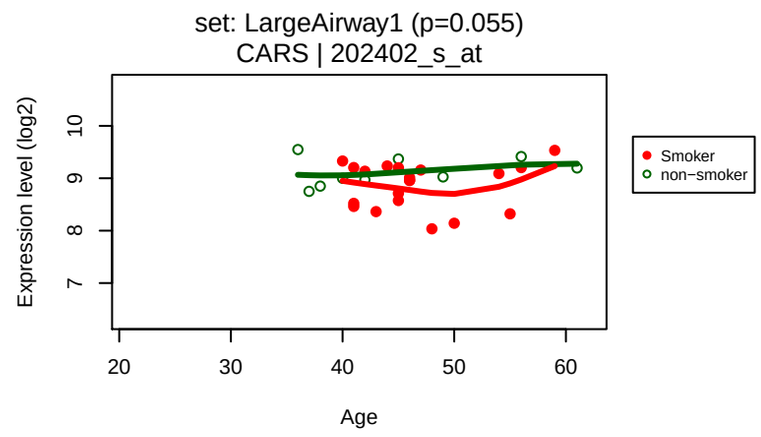

Gene expression of smokers and nonsmokers
gene: CARS | 202402_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6935
p=0.0824
p=0.0166
p=0.055
p=0.6556
p=0.1289
p=0.3557
p=0.0473
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

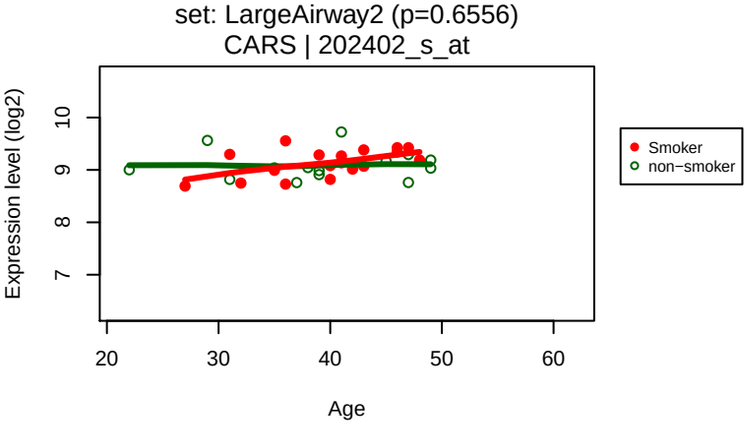

Gene expression ratio (smokers/nonsmokers)
gene: CARS | 202402_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6935
p=0.0824
p=0.0166
p=0.055
p=0.6556
p=0.1289
p=0.3557
p=0.0473
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

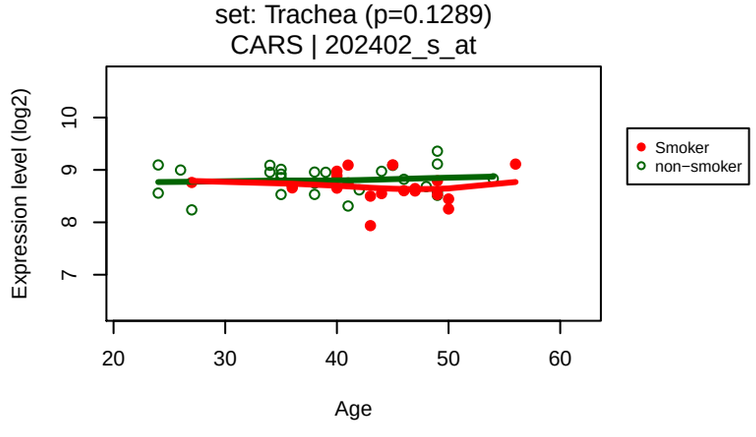

Gene expression ratio in Males
gene: CARS | 202402_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2413
p=0.0091
p=0.0794
p=0.052
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CARS | 202402_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3859
p=0.0983
p=0.7112
p=0.6477
SmallAirway2
LargeAirway2
Trachea
Macrophage2

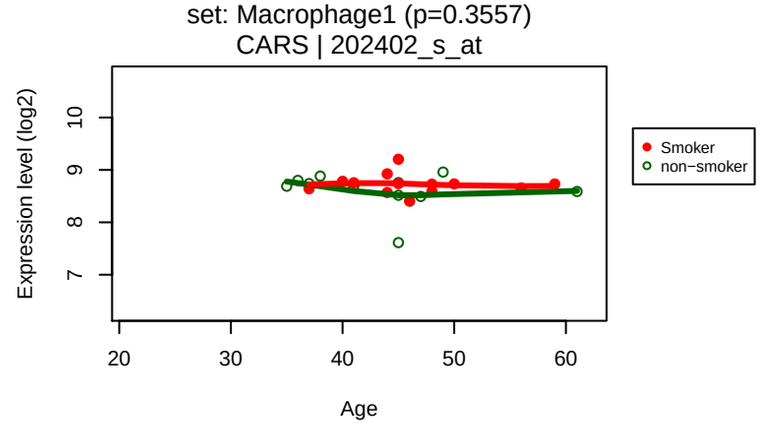

set: SmallAirway1 (p=0.6935)
CARS | 202402_s_at
Expression level (log2)
Age
Smoker
non−smoker

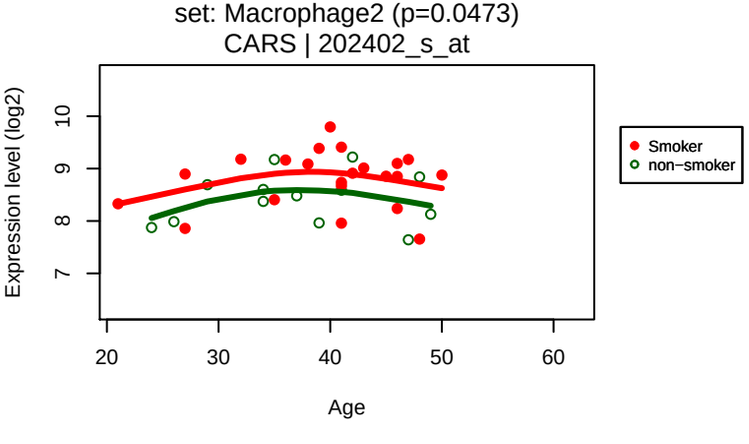

set: SmallAirway2 (p=0.0824)
CARS | 202402_s_at
Expression level (log2)
Age
Smoker
non−smoker

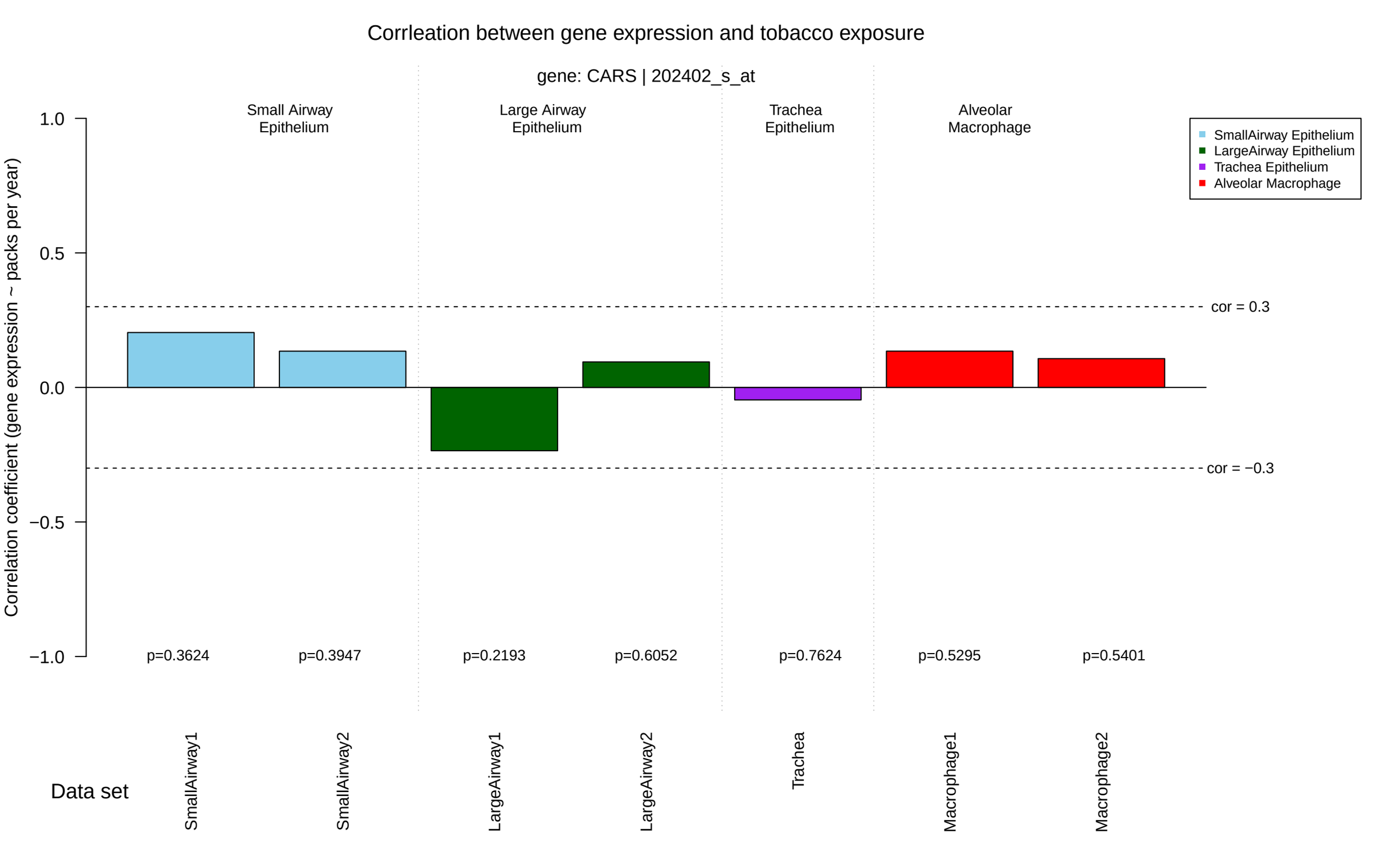

set: LargeAirway1 (p=0.055)
CARS | 202402_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6556)
CARS | 202402_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1289)
CARS | 202402_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3557)
CARS | 202402_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0473)
CARS | 202402_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CARS | 202402_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3624
p=0.3947
p=0.2193
p=0.6052
p=0.7624
p=0.5295
p=0.5401
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2